

Lab 4: IP Addressing for Secure Transmission

Introduction:

In secure networks, IP addressing plays a critical role in ensuring efficient communication and protection of data. Every device on a network must have a unique IP address. Understanding IPv4, IPv6, IP classes, subnetting, and address division is essential for configuring and securing computer networks.

Learning Objectives:

- Explain IPv4 and IPv6 addressing in detail. - Identify IP address classes and their usage.
- Calculate IP address ranges, networks, and hosts.
- Understand private IPs, subnet masks, network and broadcast addresses. - Perform IP addressing tasks for secure network design.
- Apply IP addressing concepts in lab tasks.

IPv4 Addressing:

IPv4 is a 32-bit address divided into 4 octets (8 bits each), written in decimal dotted notation (e.g. 192.168.1.1). It consists of Network Portion and Host Portion. Total addresses = $2^{32} = 4,294,967,296$.

IPv6 Addressing:

IPv6 is a 128-bit address written in hexadecimal notation, separated by colons. (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334). IPv6 was introduced due to IPv4 exhaustion. It can be shortened (e.g., 2001:db8:85a3::8a2e:370:7334). Total addresses are practically unlimited.

Classes of IPv4 Addresses

Class	First Octet Range	Leading Bits	Usage	Default Subnet Mask
A	1 – 126	0xxxxxxx	Very large networks	255.0.0.0
B	128 – 191	10xxxxxx	Medium networks	255.255.0.0
C	192 – 223	110xxxxx	Small networks	255.255.255.0
D	224 – 239	1110xxxx	Multicasting	N/A
E	240 – 255	1111xxxx	Research (reserved)	N/A

Note: 127.x.x.x is reserved for loopback (localhost).

IP Range Calculation

Number of networks = 2^n (where n = borrowed subnet bits). Number of hosts per network = $2^h - 2$ (where h = host bits). **Example:** Class C, subnet mask /26

Host bits = 6 $\rightarrow$ Hosts = $2^6 - 2 = 62$ Networks
 $= 2^{(26-24)} = 4$

Network Address & Broadcast Address

Network Address: First address in a subnet (all host bits = 0). Broadcast Address: Last address in a subnet (all host bits = 1). **Example:** Subnet 192.168.1.0/26

Network: 192.168.1.0 Broadcast:

192.168.1.63

Usable Range: 192.168.1.1 – 192.168.1.62

Private IP Addresses Class

Private IP Range

A 10.0.0.0 – 10.255.255.255

B 172.16.0.0 – 172.31.255.255

C 192.168.0.0 – 192.168.255.255

9. Default Subnet Masks Class A

$\rightarrow$ 255.0.0.0 or /8 Class B $\rightarrow$

255.255.0.0 or /16

Class C $\rightarrow$ 255.255.255.0 or /24

10. IP Division into Network and Host Portion Network

Portion: Identifies the network.

Host Portion: Identifies the device inside that network.

Example:

IP = 172.16.5.10 Mask =

/16

Network = 172.16.0.0 Host

Portion = 5.10

Lab Tasks

Task 1: Identify IP Classes & Default Subnet Masks

1. 10.45.3.29
2. 172.20.5.10
3. 192.168.1.25
4. 122.121.120.119

Task 2:

Find Network and Broadcast Address.

1. 192.168.1.100
2. 11.12.13.14
3. 10.0.5.50
4. Roll No. Roll No. Roll No. Roll No

Task 3:

Find out total number of usable networks and hosts in Class B, C & D.

Task4: Enlist all Private IP addresses range and their role in secure information systems.

Task 5 : Secure Network

By using CISCO Packet Tracer Design a complete secure network at Gulberg Campus where all four departments (CY, AI, DS, CS) can share the data easily with each-other without any data loss. Note that each user in every department uses IP address starts with your roll number. Attach screenshot of all configurations with comments.